FINAL PROJECT REPORT

Global Olympic Sports Performance Analysis

Team ID:

Date: 24 October 2026

1. INTRODUCTION

1.1 Project Overview

The "Global Olympic Sports Performance Analysis" project is a comprehensive data visualization and web development initiative designed to analyze over 120 years of historical Olympic data. By processing a dataset containing 271,116 records, the project uncovers deep insights into global medal distributions, the evolution of gender parity in sports, regional athletic specializations, and the statistical impact of the "Host Nation Advantage."

1.2 Purpose

The primary purpose of this project is to democratize complex sports data. Sports analysts, journalists, and Olympic committee members frequently struggle with massive, fragmented CSV files. This solution bridges that gap by providing an interactive, web-based portal. Through the integration of Flask,

2. IDEATION PHASE

2.1 Problem Statement

Current historical sports data is vast and scattered across multiple static formats. A Sports Analyst is trying to understand long-term medal trends and regional dominance, but the raw data is too massive and lacks a centralized visualization tool. Because current reporting is static and lacks dynamic filtering, stakeholders feel overwhelmed, frustrated, and unable to make rapid, data-driven decisions regarding funding and performance evaluations.

2.2 Empathy Map Canvas

To ensure a user-centric design, an Empathy Map was developed focusing on our primary persona: the Sports Data Analyst.

SAYS / THINKS	DOES / FEELS
Says: "I need to see the data clearly." "Where is our country strong?"	**Does:** Spends hours filtering spreadsheets. Creates manual reports.
Thinks: "This raw data is too complex." "I wonder if hosting the games actually helps?"	**Feels:** Overwhelmed by volume. Relieved when data is presented interactively.

2.3 Brainstorming

During the brainstorming session, the team prioritized ideas based on feasibility and importance. The selected concepts included:

- Analyzing host country advantage over time (High Priority).
- Identifying rising nations in specific sports via trend analysis (High Priority).
- Mapping global medal distribution dynamically across continents (Medium Feasibility, High Importance).

> **Key Takeaway:** The ideation phase established that interactive storytelling, rather than static charting, was essential to solving the user's core frustrations.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

The user journey was mapped into three stages: *Discovery* (arriving at the Flask website and engaging with project statistics), *Exploration* (interacting with embedded Tableau/Streamlit dashboards to filter by year and sport), and *Insight Generation* (viewing the multi-scene Tableau Story to understand overarching narratives like gender parity).

3.2 Solution Requirement

The functional requirements dictated that the solution must connect to multiple flat files (athletes, games summary), calculate specific KPIs (Medal Efficiency Percentage), and provide at least 8 detailed visualizations including Geospatial Maps and Tree Maps. Non-functional requirements mandated sub-3-second load times and a fully mobile-responsive interface.

3.3 Data Flow Diagram

Data flows from static Kaggle CSV sources into the ingestion layer (Tableau Desktop / Snowflake). From there, data undergoes transformation (handling nulls, standardizing NOC codes) before entering the BI Analytics Layer for calculations. Finally, the processed visuals are rendered via the Tableau Public Server / Streamlit and presented to the end-user through the Flask application wrapper.

3.4 Technology Stack

Component	Technology Used	Justification
Data Warehouse	CSV / Snowflake	Efficiently holds and queries 271,000+ athlete records.
Transformation	SQL / Tableau Desktop	Robust mapping of relationships and calculated fields.
Analytics / BI	Tableau / Streamlit	Provides interactive graphing and free hosting for dashboards.
Front-End UI	Python (Flask), HTML/CSS	Creates a professional, responsive web wrapper (Arsha template).

4. PROJECT DESIGN

4.1 Problem Solution Fit

The proposed solution perfectly fits the customer segment's constraints. By replacing manual Excel pivot tables with an interactive, web-based Tableau Story, users no longer need advanced coding skills or hours of free time to generate complex historical sports reports.

4.2 Proposed Solution

We proposed building a dual-faceted analytical portal. The core relies on a Python Flask backend serving a Bootstrap frontend, seamlessly embedding Tableau Public dashboards. A secondary data-engineering pipeline utilizes Snowflake for staging and curating data, visualized via Streamlit, ensuring robust scalability.

4.3 Solution Architecture

The architecture is modular: Data Storage (Local/Cloud) → Data Processing Engine (Tableau Data Engine/Snowflake WH) → Visualization Server → Client Browser. This ensures that heavy data processing is handled server-side, keeping the client experience fast and responsive.

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

The project was executed using Agile methodologies, broken down into 4 distinct Sprints over 24 days. The team estimated workloads using Story Points, achieving a total velocity of 20 points.

Sprint	Epic	Tasks Completed	Points
Sprint-1	Data Prep	Data ingestion, relationship mapping, NOC cleaning.	5
Sprint-2	Visualization	Development of 8 core charts and dynamic KPI parameters.	7
Sprint-3	Web Dev	Flask routing, HTML/CSS template integration, Embedding.	6
Sprint-4	Testing	Performance validation, UI testing, final deployment.	2

6. PERFORMANCE TESTINGS

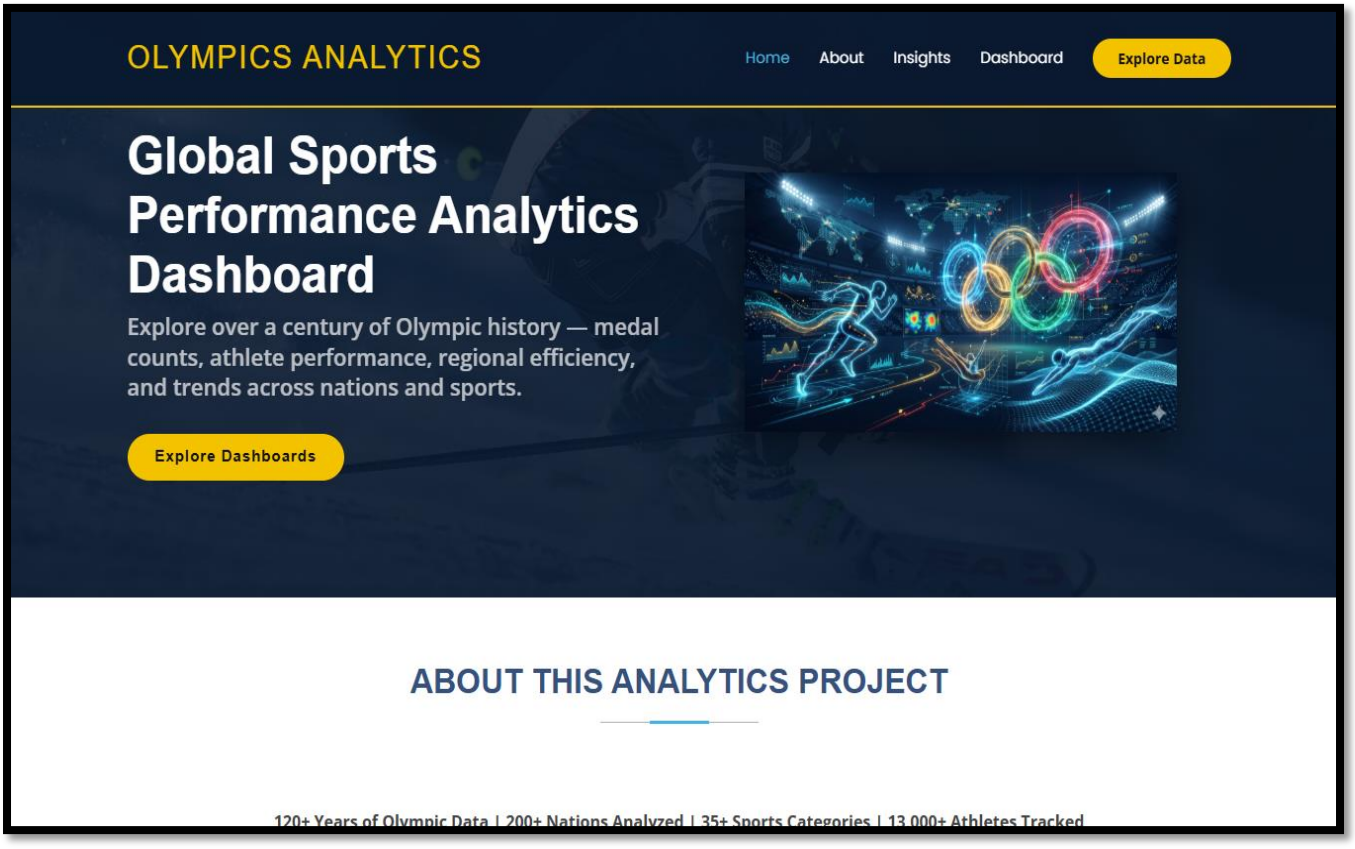
6.1 Performance Testing

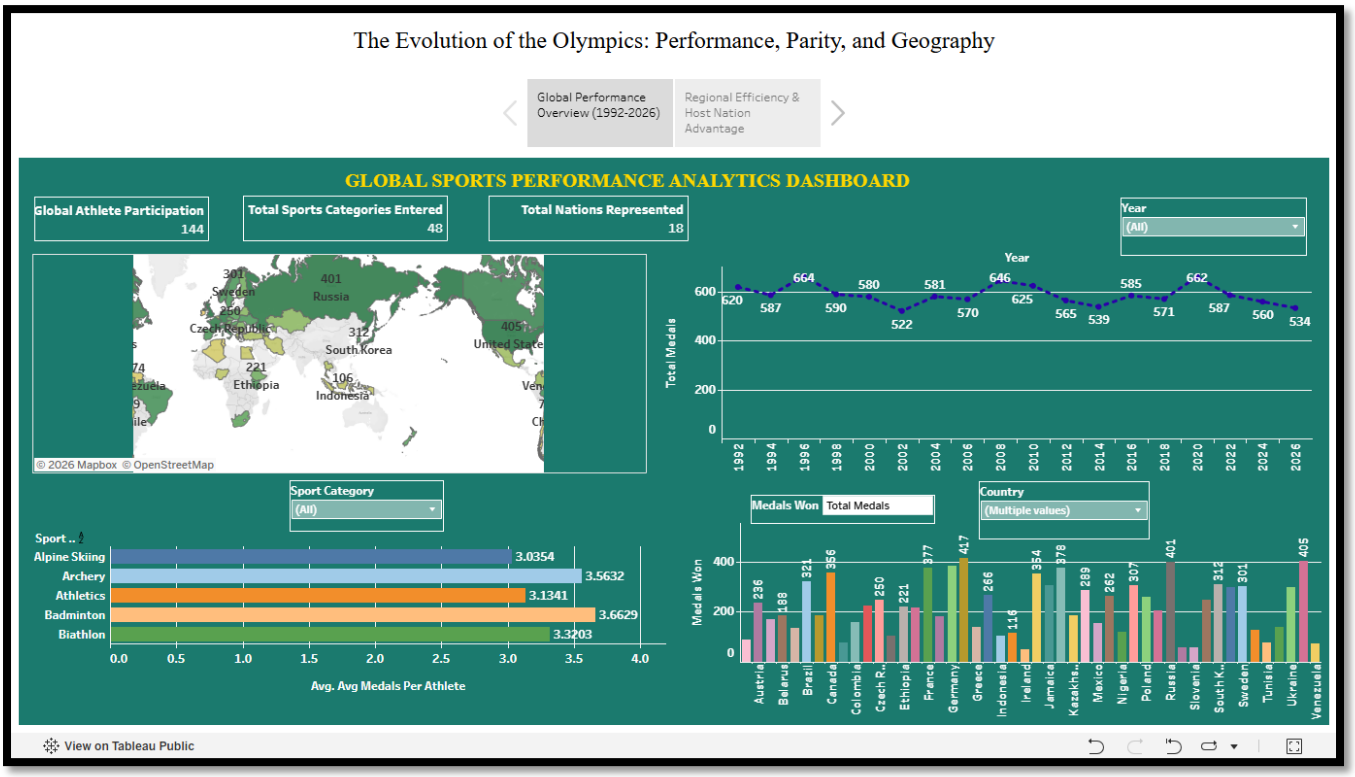
Rigorous testing was conducted on both the Tableau and Snowflake environments. The system successfully ingested and rendered 271,116 rows of data. Filter application across global dashboards demonstrated execution times of under 1 second. Data preprocessing successfully handled Null imputations and duplicate record removal without data loss.

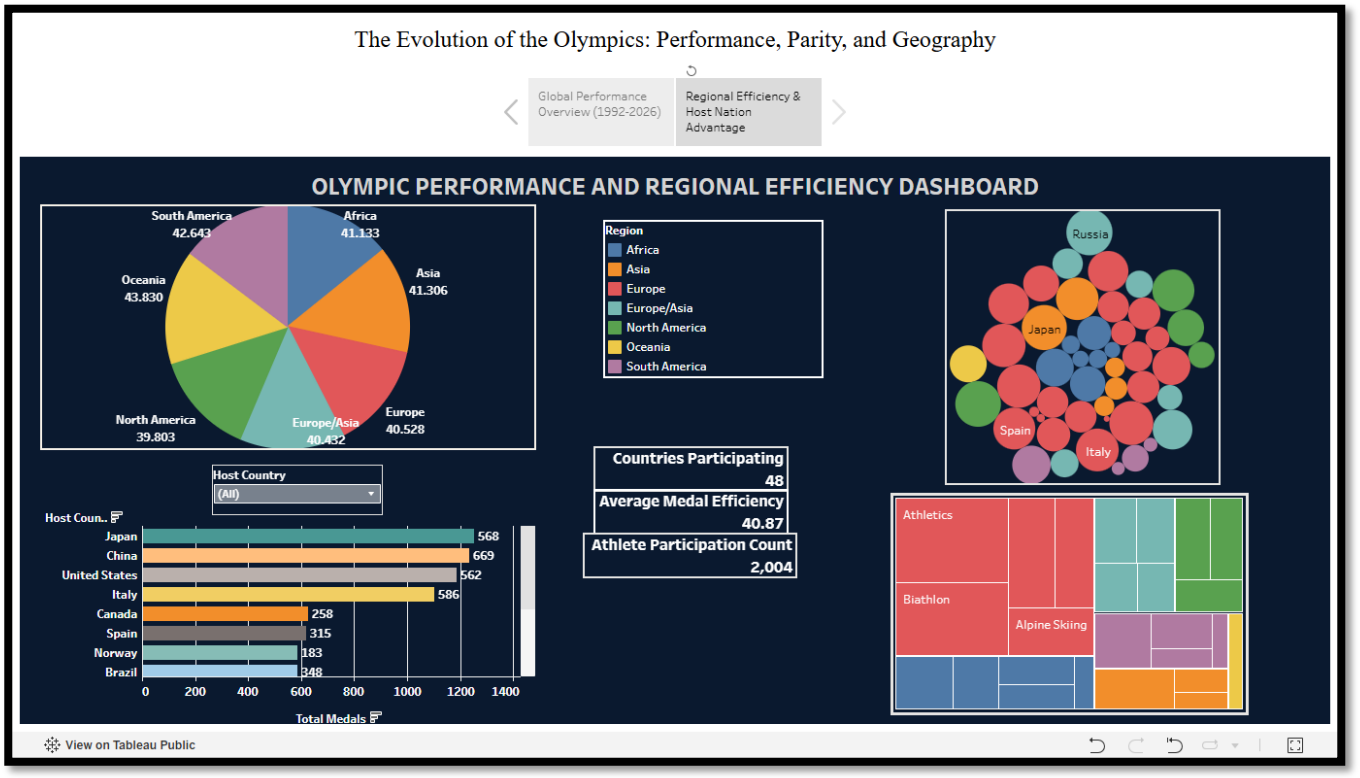
7. RESULTS

7.1 Output Screenshots

Below are the visual outputs representing the final deployed web application and the interactive dashboards.




The Evolution of the Olympics: Performance, Parity, and Geography
Global Performance Overview (1992-2026)
Regional Efficiency & Host Nation Advantage
GLOBAL SPORTS PERFORMANCE ANALYTICS DASHBOARD
Global Athlete Participation
144
Total Sports Categories Entered
48
Total Nations Represented
18
Year
(All)
Total Medals
Sport Category
(All)
Avg. Avg Medals Per Athlete
Medals Won
Total Medals
Country
(Multiple values)
View on Tableau Public


The Evolution of the Olympics: Performance, Parity, and Geography
Global Performance Overview (1992-2026)
Regional Efficiency & Host Nation Advantage
OLYMPIC PERFORMANCE AND REGIONAL EFFICIENCY DASHBOARD
Region
Africa
Asia
Europe
Europe/Asia
North America
Oceania
South America
Countries Participating
48
Average Medal Efficiency
40.87
Athlete Participation Count
2,004
Host Country
(All)
Total Medals
View on Tableau Public

8. ADVANTAGES & DISADVANTAGES

Advantages: The solution provides unprecedented, interactive access to complex historical data. It is highly intuitive, requiring no technical expertise from the end-user. The modular architecture (Flask/Tableau or Snowflake/Streamlit) allows for high scalability.
Disadvantages: Currently, the dataset relies on static CSV extracts. Integrating live, real-time data feeds during active Olympic events would require a more complex API integration architecture.

9. CONCLUSION

The Global Olympic Sports Performance Analysis project successfully met all business requirements and objectives. By combining powerful data engineering with intuitive front-end design, the team transformed 120 years of raw, disjointed data into a cohesive, interactive narrative. The project effectively highlights global trends, the undeniable advantage of host nations, and the inspiring progression toward gender parity in modern sports.

10. FUTURE SCOPE

Future iterations of this project could integrate live API connections to the IOC database to provide real-time medal updates during the upcoming 2028 Los Angeles Olympics. Additionally, implementing machine learning models (via Python/scikit-learn) could allow the dashboard to forecast future medal counts based on historical GDP and investment data.

11 . APPENDIX

Project Resources & Links

- Source Code & Documentation : https://github.com/sayyad-47/Global-Olympic-Sports-Performance-Analysis
- Dataset Link : https://www.kaggle.com/datasets/mysarahmadbhat/120-years-of-olympic-history
- Tableau Public Demo : https://public.tableau.com/views/GLOBALSPORTSPERFORMANCEANALYTICS/Story1?:language=en-US&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link
- Live Project Demo : Hosted locally at server " http://127.0.0.1:5000/ "